

TRANSPORT LAYER

The transport layer is one of the key layers in the OSI (Open Systems Interconnection) model and the TCP/IP model, which are frameworks for understanding and implementing networking protocols on the basis of TCP or UDP. Its primary role is to facilitate communication between devices across a network. Here's main functions:

- 1. End-to-End Communication**
- 2. Segmentation and Reassembly**
- 3. Error Detection and Correction**
- 4. Reliability**
- 5. Flow Control**
- 6. Multiplexing and Demultiplexing**
- 7. Connection Management**

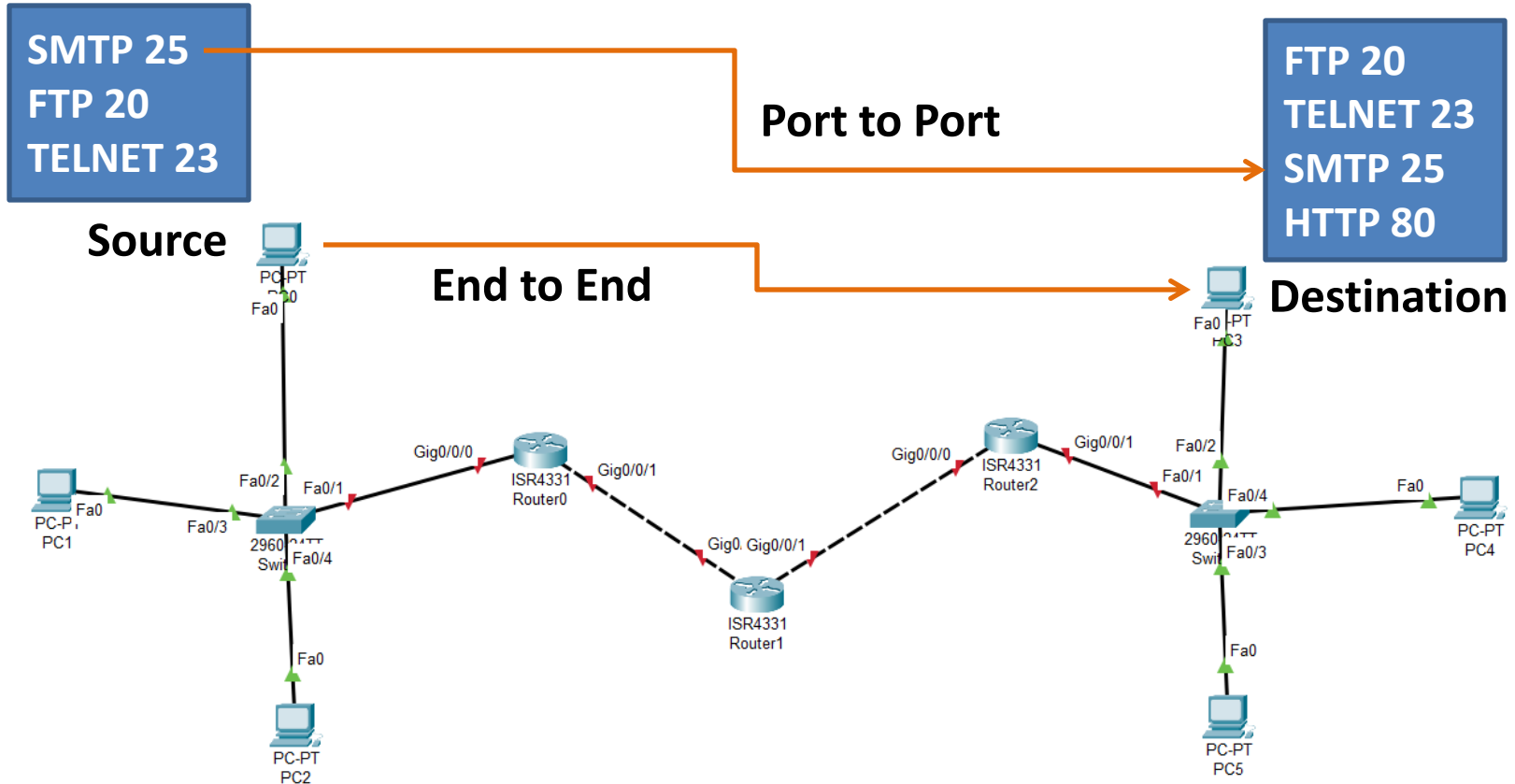
End to End Communication:

The transport layer provides end-to-end communication services for applications. It ensures that data is transferred from the source to the destination accurately and in the correct sequence.

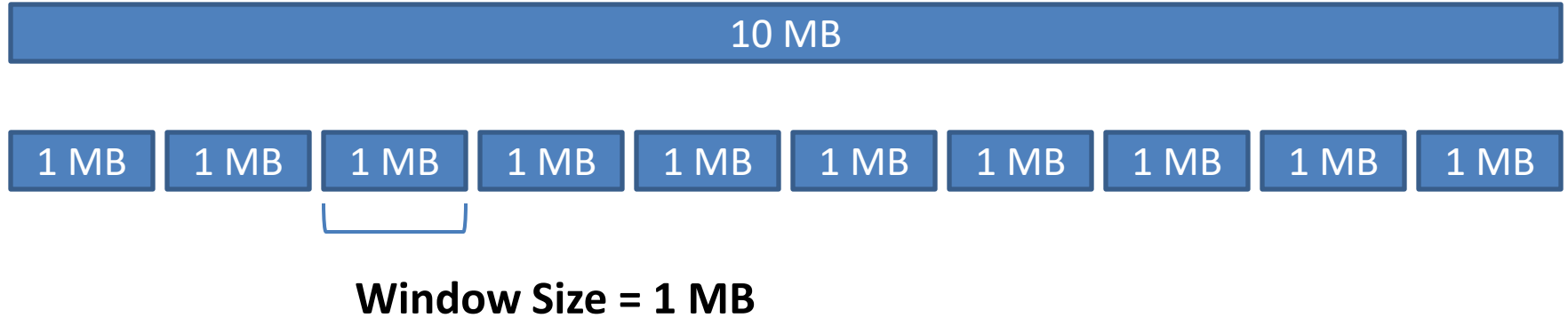
Also, provides the port to port Delivery.

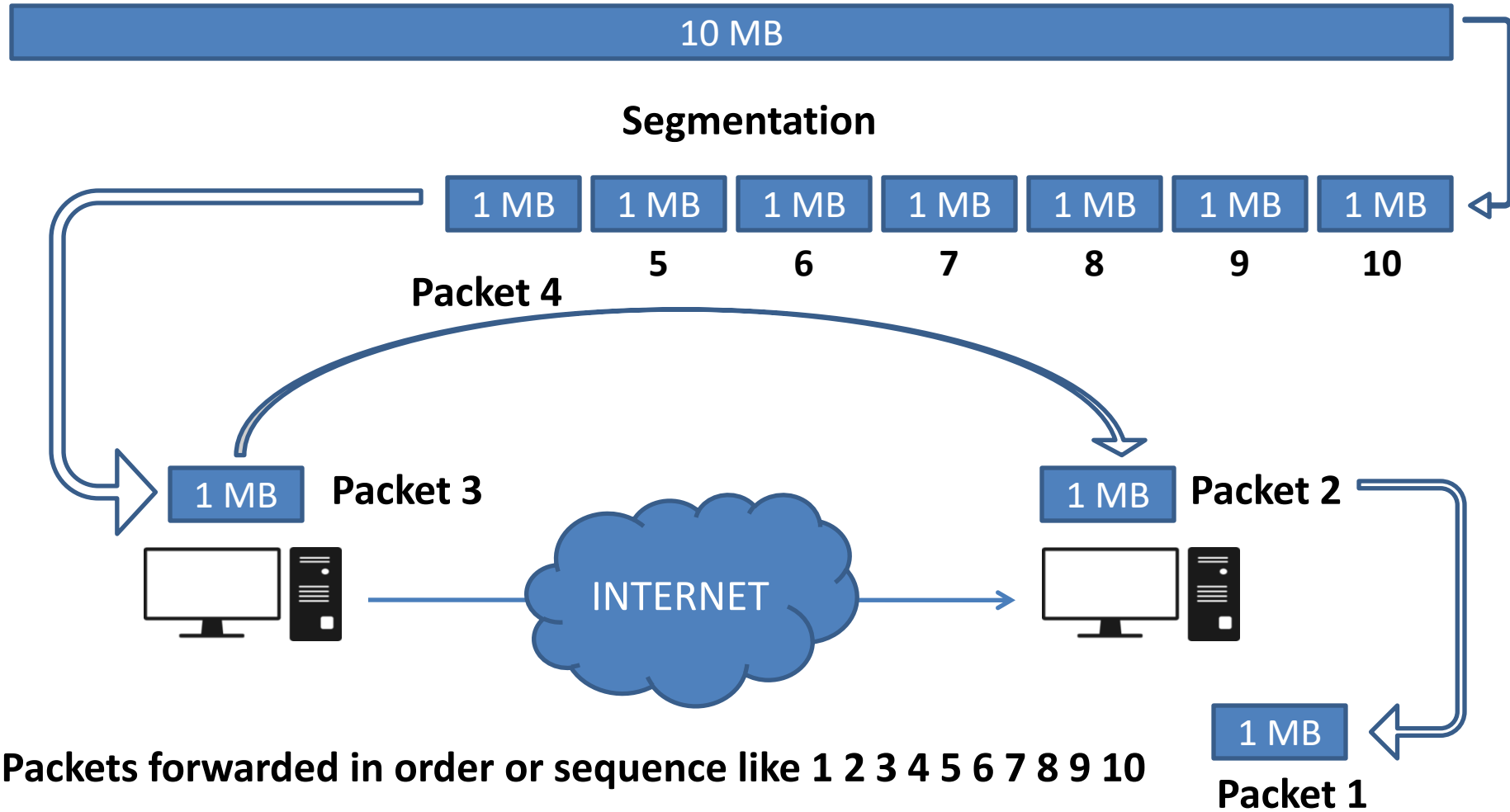
There can be multiple applications running on a Source and also on Destination so the data should be forwarded successfully as well as it must be forwarded to the specific Application and it is done by the Port to Port Delivery using Multiplexing.

End to End Communication:



Segmentation and Reassembly: When data is sent from one device to another, it is often divided into smaller units called segments. The transport layer is responsible for breaking down the data into these manageable segments at the sending end and reassembling them at the receiving end.





Reliability: One of the most critical functions of the transport layer is ensuring that data is transmitted reliably. It achieves this through mechanisms like acknowledgment of received data, retransmission of lost data, and sequencing to ensure that data arrives in the correct order.

Flow Control: The transport layer manages the flow of data between devices to prevent one device from overwhelming the other with too much data. It uses techniques like buffering, windowing, and congestion control to regulate the flow of data.

Multiplexing and Demultiplexing: The transport layer allows multiple applications to use the network simultaneously by assigning unique port numbers to each communication session. Multiplexing at the sender's end and Demultiplexing at the receiver's end ensure that data is delivered to the correct application.

Error Detection and Correction: The transport layer ensures data integrity by detecting errors in the data segments during transmission. It uses mechanisms like checksums to verify the integrity of data and requests retransmission if errors are detected.

Protocols at the Transport Layer

Transmission Control Protocol (TCP): A connection-oriented protocol that provides reliable, ordered, and error-checked delivery of data. TCP is used for applications where data integrity and order are crucial, such as web browsing, email, and file transfer.

User Datagram Protocol (UDP): A connectionless protocol that provides a faster, but less reliable, service than TCP. It does not guarantee delivery, order, or error checking, making it suitable for applications where speed is more critical than reliability, such as streaming audio and video, online gaming, and DNS queries.

TCP VS UDP

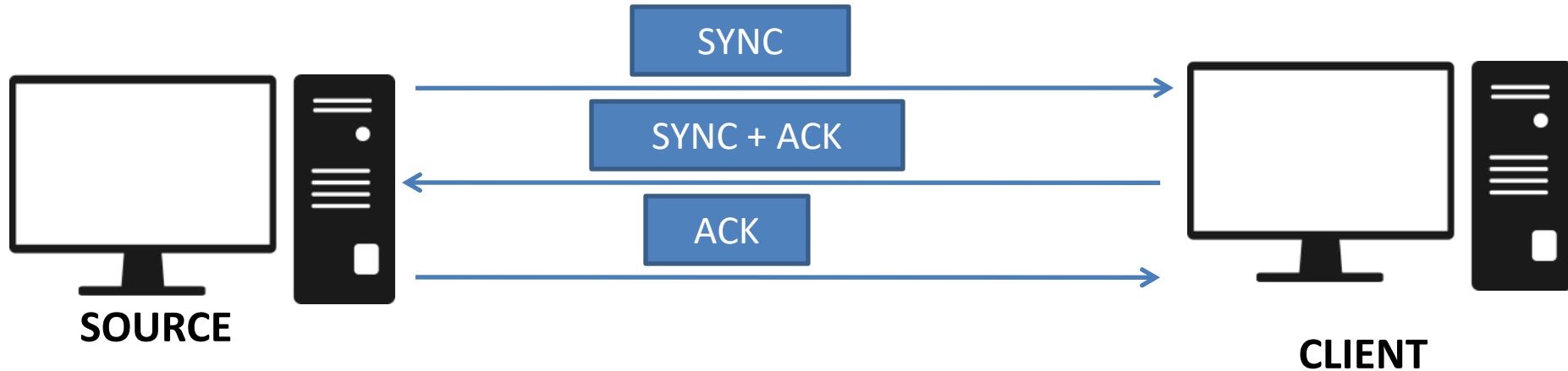
Transmission Control Protocol (TCP):

1. A connection-oriented protocol
2. Reliable
3. Slower
4. Guarantee Data Transfer
5. Used such as web browsing, email, and file transfer.

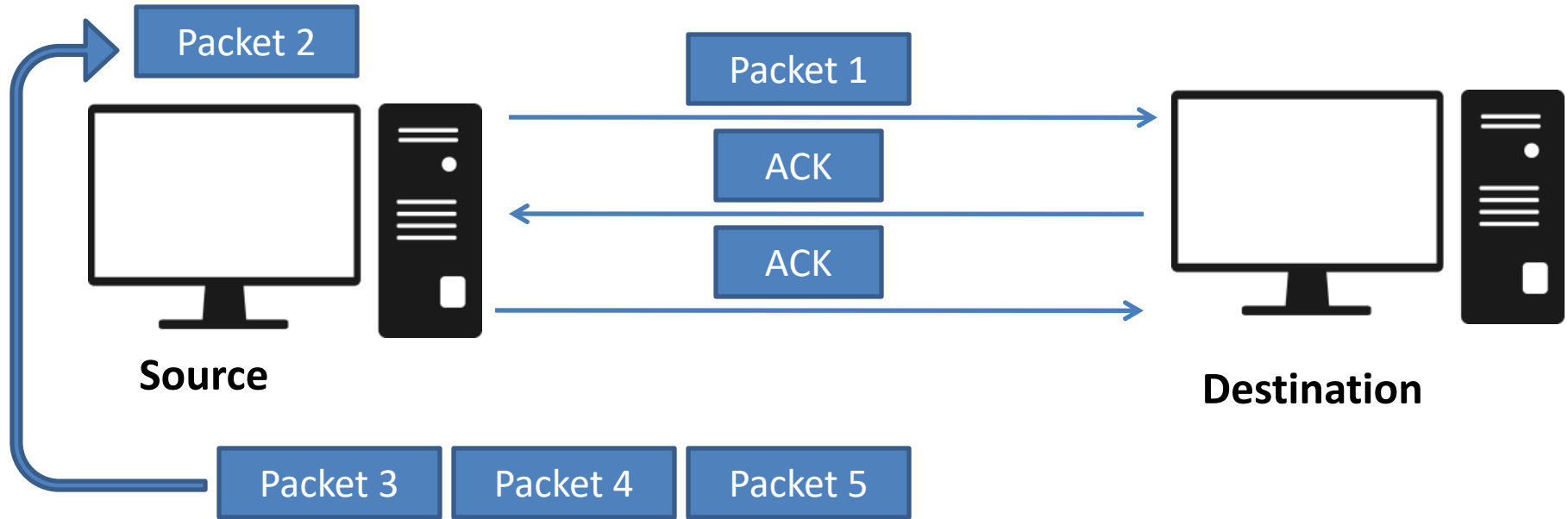
User Datagram Protocol (UDP):

1. A connection-less protocol
2. less reliable
3. Faster
4. Doesn't Guarantee Data transfer
5. Used as streaming audio and video, online gaming, and DNS queries.

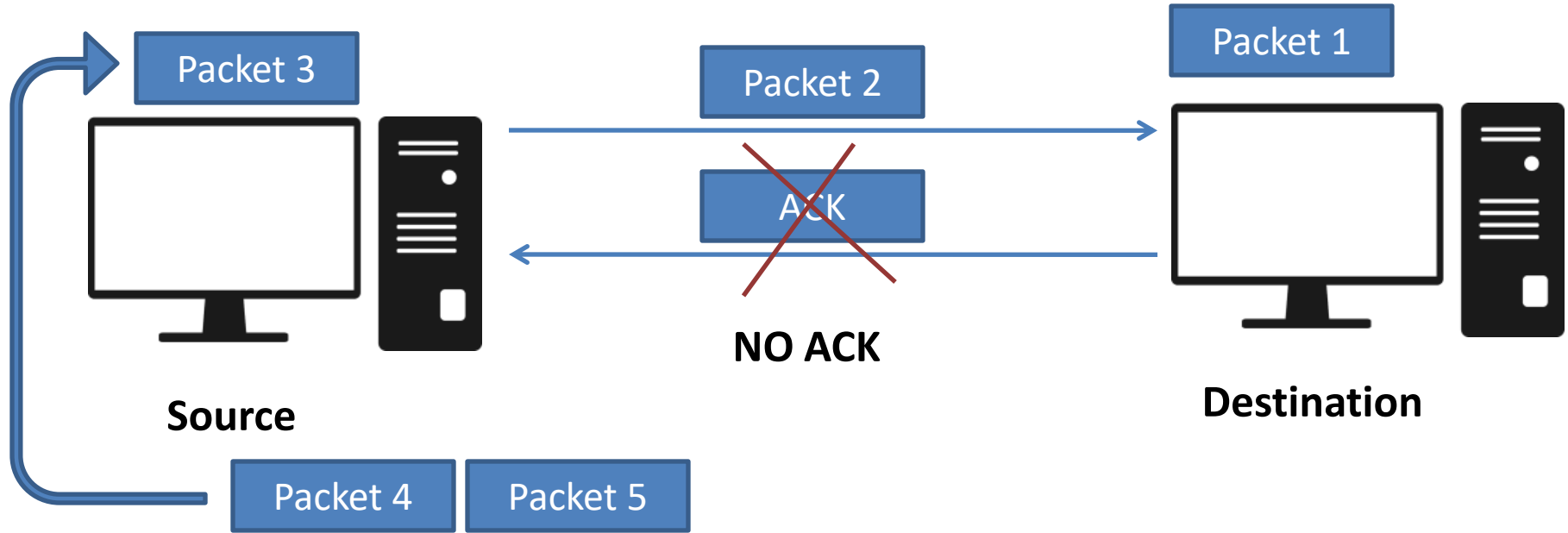
3 Way Handshake Process in TCP



3 Way Handshake and Packet Flow



UDP Protocol Don't follow the Process of 3 way Handshake



Example: Live Streaming, Voice Call, Vide Call, Internet Call etc.